Starship Cadet — Design Doc v4

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Vision & pillars

The CSV Wren is a little clockwork world: a properly simulated starship whose crew keeps it running day to day, whether you are watching or not. You arrive by shuttle as a new Works maintenance technician, and your work is part of what keeps the clockwork turning.

- **The ship runs itself.** Machines wear, supplies get used, crew follow shifts and do real work. The ship is never paused waiting for the player.
- **You are one of the crew.** The simulation treats you like everyone else: same job list, same needs, same schedule, a bunk in the junior wing like every other new hand.
- **Work is physical, not abstract.** You walk the ship, look, fetch, carry and repair. No mini-games standing in for the job.
- **Your work matters.** Fix the oven and breakfast is on time. Let the scrubber fail and the whole deck gets stuffy and short-tempered.
- **You grow by going deeper.** Progression means trust and access to the ship's more important systems, from galley ovens to the reactor.
- **Small, cozy, alive.** No combat, no fail screens. One ship you come to know like a home.

What changed from v3

V4 redesigns the ship around a daily loop across three decks and folds in everything agreed since v3 (dev notes of Oct 13–14). It stands on its own; the dev notes no longer override it.

- **Six decks, not five,** numbered from the top. The bridge gets its own small deck, and living is split into a Commons deck and a Quarters deck. Sleep on Deck 4, eat on Deck 3, work from Deck 5.
- **Crew and rooms worked out.** All 75 crew have a rank and a bed: senior suites on Deck 2, officer cabins and junior bunk rooms on Deck 4.
- **Promotion means moving.** Tier 2 brings Officer rank and your own cabin across the deck. The tier table now carries rank and room.
- **You arrive by shuttle** in the Deck 5 shuttle bay and find your own way up to your bunk.

- **Deck 5 brief** added: the ship's garage, built first so the Works bay can anchor the job list.
- **Decks as data.** The build now loads one deck at a time from a definition, with a ship-wide state that outlives it.
- **The old living deck** (build 0.3.x) is retired as a layout and kept as the style reference.
- **Kept from v3:** the clockwork, crew simulation, the job loop, the Works tiers and their gameplay, the art style, the named crew and the cozy tone.

Setting & names

You serve in the Concord Survey Service, a peaceful exploration fleet, aboard the CSV Wren: a 75-person survey ship on a long tour of a frontier shared with the Veyli, a proud, ritual-minded people.

The Veyli and the frontier stay in the background for now. They are where story events will come from later, not the main thread of the game.

Uniforms follow the classic three-colour split: red for command, gold for operations, blue-green for sciences. Departments that share a colour are told apart by their badge.

Department	Uniform	Badge	Role aboard
Command	Red	Silver diamond, gold inlay	Captain, Executive Officer, bridge officers
Works (engineering)	Gold	Silver diamond, copper inlay	Keeps every machine running. The player's department.
Watch (security)	Gold	Shield	Safety, patrols, docking, keeping the peace
Survey (science)	Blue-green	Circle	Sensors, labs, samples, charting
Infirmary (medical)	Blue-green	Cross	Crew health and rest

Ranks: Junior Officer, Officer, Department Chief, Executive Officer, Captain. The player starts as a Junior Officer in Works and can rise to Officer and senior technician, but not beyond.

The ship: six decks

The Wren has 6 decks, numbered TNG-style from the top, with a crew of 75. Its layout is a main attraction: a Works tech gets to know the hidden ship behind the walls that most of the crew never see.

Layout principles:

- **Every deck has its own shape.** The rounded hull is narrow at the top and bottom and widest at Decks 3 and 4, so each deck is recognisable at a glance.
- **Curved hull.** Walking stays on the tile grid; the outer hull follows true curves, with rounded corners and an elliptical bow.
- **Two layers to every deck.** The public ship is quarters, mess hall, labs and bridge. Behind it runs the service ship: crawlways, pipe galleries, access hatches and machine rooms.
- **Shortcuts for those who know.** Service routes cut across the ship, so learning them makes your rounds faster.
- **Places worth finding.** Quiet corners with a view, a spot where off-duty Works crew hide for tea, and odd things left in storage.

Deck	Name	Public spaces	Service layer	Size (tiles)	Opens at
1	Bridge	Bridge (sunken command ring, curved viewscreen, rear stations), ready room, observation room behind the bridge that serves as the conference room	Bridge sub-floor relays	About 14 × 10	Rare visits only, so each one feels like an occasion; relays at tier 3
2	Command & Science	Survey labs, Watch office, senior suites (Captain, XO, 4 Chiefs), guest cabin (Envoy Saruth)	Sensor and comms array, lab gas lines	To be sketched	Visitable; sensor array at tier 3
3	Commons	Large mess hall and galley, large two-level bow lounge with sweeping curved windows and a bar, infirmary, hydroponics garden	Galley machinery, cold stores, water recycler, garden irrigation	To be sketched (widest)	Start; recycler and irrigation at tier 2

Deck	Name	Public spaces	Service layer	Size (tiles)	Opens at
4	Quarters	Junior wing (8 bunk rooms, 2 washrooms, laundry, common room), officers' corridor (24 single cabins, bow nook)	Air scrubbers, heating plant, utility and supply closets	50 × 20 (widest)	Start; utility closet tier 1, scrubbers and heating tier 2
5	Operations	Works bay, tea corner, main storage, machine shop, cargo hold, shuttle bay	Air plant, fan hall, manifold room, battery room, pipe galleries, cargo lift machinery	44 × 16	Works complex at start; machine rooms tier 2; battery room tier 3
6	Engineering	Engine room overlook	Reactor core, drive coils, power junctions, the lowest crawlways	To be sketched	Tier 3 and 4

The daily loop spans three decks: sleep on Deck 4, eat on Deck 3, work from Deck 5. The lift is part of the core from the first playable build.

The spine. A vertical core runs through all six decks with the main lift, a maintenance ladder and the main power and air trunks. Service crawlways branch off it on every deck. Crawlways have low ceilings, so you crouch-walk through them. A freight lift joins the Deck 5 cargo hold to Deck 3's cold stores.

Build order: Deck 4 (built), then Deck 5, then Deck 3, then 2, 6 and 1 as progression opens them.

Crew and rooms

All 75 crew have a rank and a bed, and where you sleep shows where you stand: senior crew in suites on Deck 2, officers in single cabins and juniors in bunk rooms on Deck 4.

Rank	Count	Lives in
Captain	1	Suite, Deck 2
Executive Officer	1	Suite, Deck 2
Department Chief	4 (Works, Watch, Survey, Infirmary)	Suites, Deck 2

Rank	Count	Lives in
Officer	22	Single cabins, Deck 4 (24 built, 2 spare)
Junior Officer	47	Bunks, Deck 4 (8 rooms × 6 = 48 berths, 1 empty)

Deck 4, the Quarters deck. The junior wing sits aft of the lift: 6-berth bunk rooms, two shared washrooms, a laundry and a junior common room with stern windows. The officers' corridor runs forward of the lift toward the bow: single cabins, carpet, warm lamps and quiet after 23:00, ending in a bow nook. Twelve cabins have hull windows; twelve are inner cabins.

The night-shift room. One junior bunk room houses the night shift, so it is dark and hushed at midday.

Promotion means moving across the deck. The player starts in a junior bunk room (Bunk 1, with Tomas; Pell and Rina are next door in Bunk 2). Promotion to Officer earns a cabin in the officers' corridor, and walking it for the first time with your own key is the moment of promotion.

Named crew so far: Captain Odile Marrow, Commander Teo Vance, Dr. Imra Sato, Pell Okafor, Rina Deshi, Chief Bram Holt, Mother Fen, Tomas Lind and Envoy Saruth (Veyli), who has the guest cabin on Deck 2. The game also needs a Works Chief, your boss and the one who certifies you, still to be named and designed; they live in a Deck 2 suite and work from a glass office on Deck 5.

Deck 5: the working deck

Deck 5 is the ship's garage, not its home, and it is built next so the Works bay can anchor the shared job list. Where Deck 4 is homey, Deck 5 is a workshop: steel plate and grating floors, gold-and-charcoal hazard striping at doorways and machines, pipe runs along the walls, bright cool work lights and very few windows. The plan is in deck5-plan.html (sketch 2).

Shape. About 44 × 16 tiles, narrower than Decks 3 and 4. The lift lines up with Deck 4's, a little aft of midships, splitting the deck into a cargo end aft and a working end forward.

Area	What's there	Opens at
Shuttle bay (stern)	One small shuttle on its pad; the flat stern hull is the bay doors; a control booth with glass over the pad. Where you first arrive.	Start (bay floor)
Bay door gear	Giant gears and rams in the stern corner; the doors slide into a pocket here. Glass panel onto the pad.	Sealed, visible through glass
Cargo hold	Crates off the shuttles plus the big spares, and the freight lift up to Deck 3's cold stores. Where supplies enter the ship.	Sealed until step 7
Haulway and lift lobby	The walkable route from the shuttle pad to the lift, opening straight into the Works bay.	Start
Works bay	Big job board (the shared job list's home), workbenches, tool wall, kit lockers, a window onto the shuttle pad, and the Works Chief's glass office overlooking the floor.	Start
Tea corner	Behind the parts racks, where off-duty Works crew hide. One warm lamp.	Start
Main storage	Aisles of shelving sorted by system (lights, filters, plumbing, galley parts) and a check-out counter. A dusty back corner of things nobody remembers ordering. You carry two items, so aisle layout shapes routes.	Start
Machine shop	Lathe, parts fabricator, welding bench. When storage runs out of a part, make one: slower than fetching.	Tier 1
Battery room	Rows of cells with glowing charge bars.	Tier 3
Air plant (bow)	Oxygen cells in glass columns, scrubber stacks.	Tier 2

Area	What's there	Opens at
Fan hall (bow)	The ship's lungs: one huge slow fan, ducted under the main corridor to the spine trunk. Visible through a window at the corridor's end from day one.	Tier 2
Manifold room (bow)	A wall of colour-coded pipes and valves serving every deck; where tier 2 fault tracing starts.	Tier 2
Pipe galleries	Narrow crawls along both hull sides, storage to the air plant and machine shop to the manifold. Later shortcuts, and part of tracing connected faults.	Tier 2

First build of Deck 5 (step 6c): lift lobby, haulway, corridor, Works bay, tea corner, washroom, storage, machine shop and the shuttle bay floor, plus the glass onto the pad and into the fan hall. Everything else is sealed with tap text, as on Deck 4.

The clockwork: systems, supplies and the job list

Every machine aboard has a condition that wears down with use and time, and wear turns into work on one shared job list that the whole crew draws from.

Machines. Each one is a real object in a room, with a condition from 100% to 0%, a wear rate and the effect it has on the ship when it falters. Wear depends on use: the galley oven wears fastest around mealtimes, the laundry washers on the morning after a busy shift, the lift whenever people ride it.

A machine's life runs through five states:

1. **Healthy.** Works normally.
2. **Worn.** Still works, but a sharp-eyed tech can spot it. Servicing now is cheap and quick.
3. **Faulty.** A fault is logged on the job list. The machine works badly: slower, noisier, less effective.
4. **Failing.** Effects spread to the crew and the ship. The job is marked urgent.
5. **Broken.** The machine stops. Repairs take longer and need more parts.

Supplies. Supplies come aboard by shuttle into the Deck 5 cargo hold and are shelved in main storage: filters, spare parts, tools, linen and food stores (sent up to Deck 3's cold stores by the freight lift). Repairs use parts and cooking uses food. Low stock adds restock work. An empty shelf blocks a repair until supplies arrive, unless you make the part in the machine shop, which is slower.

The job list. One list for the whole ship, shown on the big job board in the Works bay and on your wrist comm. The ship adds jobs itself: faults, routine servicing, restocking, cleaning, cooking. Crew on duty claim jobs that fit their department and skill, walk to them and do the work. Jobs nobody claims pile up and get worse.

Consequences. Machines affect crew needs and mood, often on another deck:

Machine	When it falters
Air scrubber	Air gets stuffy, energy and mood drop on that deck
Water recycler (Deck 3)	Showers run weak on Deck 4, hygiene and mood drop
Laundry washers	Crew go about in grubby uniforms, mood drops
Heating	Cold cabins, grumbling crew
Galley oven	Meals run late, hunger rises, Mother Fen gets stressed
Lights	Rooms dim, people move slower and grumble
Doors and lift	Routes get longer, people run late for shifts

Uniforms are objects that get grubby. You drop yours at the laundry and take a clean one from your locker.

Crew simulation

Every crew member is a simulated agent with a job, a shift, a bed, needs and opinions of the others, and their day is decided by the same rules as yours.

Shifts. Two shifts: a busy day shift and a much quieter night shift. Most of the crew work days, so the ship has a real daily rhythm. Nights are dim and calm, with a skeleton crew keeping watch.

Shift	Share of crew	On duty	Off duty	Asleep
Day	About 80%	08:00–17:00	17:00–23:00	23:00–07:00
Night	About 20%	21:00–06:00	14:00–21:00	06:00–14:00

During the night, only urgent jobs and essential systems get attention. Routine work waits for morning, which is when the job list is busiest.

A day moves through three decks. Crew wake on Deck 4, eat on Deck 3 and work wherever their department is, so the lift and the lobbies have rush hours.

On duty, crew claim jobs from the job list that match their department and skill, walk there and do the work. A Works tech fixes the scrubber; Mother Fen cooks; Infirmary checks on whoever is run-down.

Off duty, each person picks what to do by scoring their options against their needs and traits: eat, sleep, wash, socialise, relax in a favourite spot, a hobby.

Needs. Hunger, energy, social and hygiene, from 0 to 100. Showers restore hygiene, which gives the washrooms a morning rush. Mood comes from needs, the state of the ship and the people around them.

Relationships. Each pair of crew has an affinity that drifts as they eat, work, bunk and spend time together. Friendships and small rivalries form on their own.

Skill. Each crew member has a skill level in their department. Skilled crew work faster and cause fewer follow-up faults.

Performance. Only crew on the deck you are viewing walk tile by tile. Everyone else is simulated at room level, so the full crew of 75 stays cheap on a phone. A busy deck shows about 15–25 people at once.

The first crew build uses Decks 4, 5 and 3 with about 30 crew. The full crew of 75 (about 60 on days, 15 on nights) arrives as decks are added. About 10 are hand-written story characters; the rest are generated with names, looks, jobs and personalities.

The player: Works technician

You work the day shift as a Works maintenance tech, taking jobs from the same list as the rest of the crew and doing them with your own hands.

Arriving by shuttle. The game opens with your shuttle landing in the Deck 5 shuttle bay. You step off onto the pad, walk the haulway to the lift lobby, glimpse the Works bay that will be your workplace, and find your own way up to your bunk on Deck 4. Until the arrival is built, the game starts with you waking in Bunk 1. Still to decide: which milestone builds it, and who meets you off the shuttle.

The job loop. Every job follows the same physical steps:

1. **Notice.** A fault arrives on your wrist comm or the Works bay job board, someone flags you down, or you spot a worn machine on your rounds.
2. **Diagnose.** Open the panel and inspect it. You see which component is failing and what the fix needs.
3. **Fetch.** Get the part or tool from main storage or the Works bay, or make it in the machine shop. You can carry two items at a time.
4. **Repair.** Hold to work. Repairs take longer on broken machines and go faster as your skill grows.
5. **Done.** The machine recovers, the job closes, and the people it affected notice.

What makes it interesting:

- **Priorities.** Several jobs compete. The urgent scrubber, the oven Fen keeps asking about, or the routine service that prevents next week's breakdown?
- **Prevention.** Spotting wear early is cheap and quick. Waiting for faults costs more time, parts and goodwill.
- **Routes.** Walking matters, and the job spans decks: parts are on Deck 5, the scrubbers on Deck 4, the oven on Deck 3. Fetching the right parts in one trip saves real time on your shift.
- **Interruptions.** The ship keeps living around you. People stop you for help, argue in corridors, or ask you to join them at lunch.
- **Reputation.** Crew remember who keeps their corner of the ship running. Good work raises friendship and trust.

No hard failure. If you slack, other Works crew pick up jobs, but more slowly, and your supervisor notices. The ship never dies because of you, but it gets visibly worse.

Off shift, you are free: eat, wash, socialise, explore, sleep. Some crew will ask for small favours, like fixing a lamp in their cabin, which build friendships and earn goodwill.

Thirty seconds of play. Your wrist comm chirps: scrubber 2 on Deck 4 is at 60%. You head out of the junior wing, passing Rina and Pell arguing in the corridor. You open the scrubber panel; it needs a filter, so it's down the lift to storage on Deck 5. On the way back, Fen calls: the oven on Deck 3 is acting up again. Now, or after the scrubber? You choose, and the ship reacts.

Works progression: deeper ship systems

You progress by earning clearance to work on deeper, more connected systems, and each tier changes how the job plays, not just how big the numbers are. Tier 2 is also your first promotion: Officer rank and your own cabin.

Tier	Title	Rank	Room	Systems you can work on	What's new in the gameplay	Parts of the ship opened
1	Junior Technician	Junior Officer	Bunk, junior wing	Galley oven, lights, doors, washroom fixtures, laundry washers, cabin fixtures	One machine, one fault. Learn the loop: diagnose, fetch, repair, service	Works bay, storage, machine shop, utility closets
2	Technician	Officer (first promotion)	Own cabin, officers' corridor	Air scrubbers, water recycler, heating, air plant, fan	Connected systems. Ducts and pipes run between rooms and decks, so a fault can start upstream. Trace it from the manifold room and junctions	Air plant, fan hall, manifold room, pipe galleries, Deck 4 scrubbers and heating, Deck 3 recycler
3	Senior Technician	Officer	Own cabin	Power conduits, relays, batteries, sensor array	Planning around the crew. Some repairs need a shutdown. Pick a quiet hour, warn affected crew, and decide which systems get power during a fault	Battery room, power junctions, crawlspaces, bridge sub-floor

Tier	Title	Rank	Room	Systems you can work on	What's new in the gameplay	Parts of the ship opened
4	Engine Specialist	Officer	Own cabin	Reactor, drive coils, engine controls	Team jobs. Big repairs need several techs. You plan the job, call in crew and coordinate, often timed with the bridge	Deck 6 engine room

How you earn a tier:

- **Experience.** Jobs completed in your current tier, weighted by quality and how early you caught faults.
- **Certification.** When you are ready, the Works Chief sets a supervised qualification job in the next tier. Pass it and your clearance goes up.
- **Trust.** Reliability matters: turning up for shifts, not leaving jobs half-done, keeping your corner of the ship running.

What each tier gives you:

- **Access.** Keycards to new rooms and systems. The ship literally opens up as you grow, and at tier 2 your key opens a cabin of your own.
- **Tools.** A diagnostic scanner, a pipe kit, a conduit tester, a calibration rig. Each one makes new kinds of faults readable.
- **Responsibility.** Night on-call duty, where emergencies can wake you. Later, a junior tech to mentor.
- **Skill.** Repairs get faster and cleaner in tiers you have mastered, so old jobs become quick wins while new ones stay challenging.

Why it stays interesting: each tier adds a new kind of thinking. Tier 1 is fixing things, tier 2 is tracing problems, tier 3 is planning around people, and tier 4 is leading a team.

Later: other jobs and the story layer

Both come after the Works sandbox works. Version 1 is Works only.

Other jobs and transfers. Other departments come later as transfers, all drawing from the same job list, so each one is mostly new jobs and tools rather than new systems. Each would need its own progression ladder, like Works has.

Job	Core activity	Ties into the sim through
Galley / crew services	Cook from the stores, learn favourites, time meals to shift changes	Hunger, mood, food supplies
Infirmary aide	Notice who's run-down, treat them, send them to rest	Energy, health, mood
Survey lab assistant	Collect sensor readings, run samples through lab equipment	Sensors, story threads

Story layer. Story sits on top of the sandbox, so events grow out of a living ship rather than replacing it.

- **Crew events.** Small moments the sim can trigger: a friendship scene, an argument that needs smoothing over, a birthday in the bow lounge.
- **Ship events.** Bigger situations that put the systems under pressure: a power surge, a Veyli vessel needing aid, a sensor anomaly.
- **Special missions.** Startable when you're ready, shown as an icon on the job list or ship map. Occasionally an away job by shuttle, like repairing another ship's systems.
- **Character arcs.** The named crew have personal threads that open up as friendships grow.
- **The bridge.** Deck 1 is rarely visited, so story events are the natural reason to go: a briefing in the observation room, a repair timed with the bridge.

Art, camera & controls

The look is locked: textured low-poly in a cozy isometric cutaway, with every texture drawn in code. The old living deck from builds 0.3.x is kept in the build as the style reference.

- **Camera:** orthographic isometric view that follows you. Walls between you and the camera drop to knee height, a straight run at a time; tall furniture fades a room at a time. Rotate in 90° steps; pinch or scroll to zoom.
- **Materials:** procedural textures (deck plating, carpet, wall panels, fabrics, wood, rugs, screens), soft shadows and warm lamps. Lighting shifts from cool white by day to amber at night.

- **Each deck has its own character.** Deck 4 is homey: carpet and warm lamps in the officers' corridor, plate floors and work lights in the junior wing. Deck 5 is a workshop: steel plate, grating, hazard striping, pipe runs and cool work lights.
- **Curved hull:** the deck keeps its tile grid for walking, but the outer hull follows the ship's curved shape, with rounded corners and an elliptical bow. Windows are openings in the curved hull. Inner glass walls arrive with Deck 5.
- **Characters:** built from a one-line spec with skin tone, hair style and colour, height, build, accessories, department colour and badge. Walk animation with planted feet.
- **Controls:** tap to walk to a spot or object, press and hold to walk toward your finger, tap a person to talk. Tap a lift door for a panel of all six decks. Doors slide open for anyone nearby; sealed rooms give tap text.
- **Crawlways** get low ceilings and a crouch-walk.
- **UI to come:** wrist comm with the job list, machine inspection panel, a two-slot carry indicator, and watch tools (time speed, crew panel, systems panel).

Technical plan: how decks work in the build

The game stays one HTML file using three.js r128 from cdnjs, built in chat on a phone (Galaxy TriFold), with no image assets. Ship, decks, machines, crew and jobs are all stored as data, so adding content never means rewriting code.

One deck in memory at a time. Only the deck you are on is built. Leaving a deck unloads it completely; built decks are never cached, because a single floor texture is 4096 × 2048.

Ship state outlives decks. A ship-wide state (`SHIP`) holds everything that must survive a deck change. A deck is rebuilt from its definition plus that state, so leaving a deck loses nothing. Today it holds unlocked doors per deck; machines, supplies and wear join in step 7.

Deck definitions. Each deck is one block of data in `DECKS` : number, name, random seed, lift position, a text map of rooms, doors, hull shape, windows, sealed rooms and their tap text, and a `dress` function that furnishes it in a fixed order. The seed means a deck comes out the same every time. Deck plans are sketched as stand-alone HTML in the same format (deck5-plan.html drops straight into `DECKS[5]`).

Shared kit vs. current deck. Materials, textures, furniture builders and room templates are made once at load and shared by every deck. Everything a deck build creates belongs to that deck and is freed on unload.

Room templates. A room is furnished from its size and where its door is (bunk room, washroom, laundry, officer cabin so far), and the same template works whichever way the room faces. Templates also record the spots people use: beds, showers, basins, washers, desks.

Ship frame. Every deck is placed so its lift sits at the same world position, which keeps the spine lined up from deck to deck. Freight lift platforms are aligned the same way (Deck 5′s must meet Deck 3′s cold stores).

Changing decks. `goToDeck(n)` hides the crew, unloads, builds deck n, puts you by the lift and brings in the crew who are on that deck. Each crew member has a deck; only those on yours walk tile by tile. The lift′s fade covers the rebuild.

Rules that keep it working on phones:

- Never let two surfaces share a plane: overlap solids, or keep faces at least 0.005 apart.
- Keep the canvas opaque, the camera depth range tight, and snap the camera to whole pixels.
- Avoid per-pixel noise on large surfaces; it shimmers on high-resolution screens.
- 1 unit equals 1 floor tile. Furniture blocks tiles; walls sit on the edges between rooms.
- Decks with different lamp counts may compile extra shaders on first visit. If the lift′s fade doesn′t hide the pause, pad every deck to the same number of point lights.

Testing. cadet_test.py compares two builds headlessly (Playwright and SwiftShader) with a virtual clock, so runs are deterministic: deck data, paths, collisions, crew positions, taps and screenshots. Real-device checks on the TriFold are the final word.

Working method. The design doc, latest build, deck plans, cadet_test.py and the dev notes live in a Claude Project. Builds are edited in place, never rewritten whole. Turns stay short, each ending with a tested build handed over as a file and downloaded straight away. When a chat gets long, the dev notes are updated as a handoff for a fresh chat.

Milestones

The current build is 0.5.0 (about 2,440 lines): Deck 4 built from deck data, ready for other decks to be added and switched between. Next up is step 6c, Deck 5 and the lift.

Done so far:

Build	What it added
0.4.2 – 0.5.0	Step 6a: decks as data, unloading, ship frame, crew decks, ship state. Plays exactly as 0.4.1. Deck 5 planned (6b).
0.4.1	Step 5: Deck 4 finishing touches
0.4.0	Step 4: Deck 4 junior wing; officers' wing built and locked
0.3.4	Step 3: curved hull walls
0.3.3	Step 2: room templates
0.3.2	Step 1: kit carry-over, wall runs, v3 uniforms and badges
0.3	The living deck (now the style reference), press-and-hold walking
0.2	Character kit and walk animation
0.1	Art style test, one cabin

Milestone 1, Clockwork core (remaining steps):

1. **6c. Deck 5 and the lift**, one tested build per turn:
   a. Deck 5's shell straight from the plan: new floors (hazard-striped plate, grating), inner glass walls, sealed rooms with tap text.
   b. The lift: tap a lift door for a panel of all six decks, unbuilt ones greyed out. A ride closes the doors, fades, changes deck and opens them again.
   c. Dress the Works complex: job board, benches, tool wall, kit lockers, tea corner, washroom, storage aisles, machine shop.
   d. The shuttle bay and the view: shuttle on its pad, bay doors, door gear behind glass, the fan turning in the fan hall. Some Works crew on Deck 5 (Pell first).
2. **6d. Deck 3:** sketch its plan (the freight lift must meet its cold stores), then build its starter rooms (mess hall, lounge).
3. **7. Machines, supplies and the shared job list,** anchored on the Works bay job board.

Later milestones, each playable and tested in chat:

2. **Crew on shifts.** About 30 crew on day and night shifts with four needs and routines across Decks 3–5. On duty, they claim and do jobs. Watch tools: time speed, crew panel, systems panel. Test: watch a full ship day and check it feels alive.
3. **You as a tech.** Wrist comm, inspect, carry, repair. Tier 1 jobs on Decks 3–5. The shuttle arrival, if not built earlier.
4. **Consequences and reputation.** Machine effects on crew needs and mood, uniforms and laundry, and crew remembering who helped.
5. **Progression to tier 2.** Certification, promotion to Officer and your own cabin, the Deck 5 machine rooms and connected-system faults.
6. **The rest of the ship.** Decks 2, 6 and 1, the spine's crawlways and pipe galleries, building toward all 75 crew.

Open questions

- Who is the Works Chief? Name, personality and how they judge your work. (Lives on Deck 2, works from the glass office on Deck 5.)
- Who meets you off the shuttle, and which milestone builds the arrival?
- How long is a ship day in real time? This sets every wear rate; 10 minutes was the old guess, and a working shift may need more.
- Should the player ever move to the night shift, for example as part of on-call duty?
- How does diagnosis look on screen: a simple panel with labelled components, or something more hands-on?
- Can the player be passed over for certification, or only delayed?
- Should the player customise their character at the start?
- What triggers the first story events once the sandbox works?
- Does the diplomacy-and-story starship game become its own future project?

Answered since v3: the player starts in Bunk 1 with Tomas, and Pell and Rina are in Bunk 2.